Active Directory Lab

Project Overview:

Built a virtualized Active Directory environment using Windows Server 2022 and Windows 11 virtual machines to simulate enterprise IT infrastructure. This lab was designed to replicate a real world help desk workflow and responsibilities including user management, password resets, group membership administration, computer domain joining, and Group Policy implementation.

Lab Environment & Tools:

Environment	
Component	Details
Hypervisor	Oracle VirtualBox
Server OS	Windows Server 2022
Client OS	Windows 11
Domain Name	charlie.local
Host Name	FL-DC-01
Client Hostname	Desktop01

Tools	
Tools	Purpose
Active Directory Domain Services	Domain, user, and security management
Windows Server Manager	Server and role administration
Active Directory Users and Computers	User and account administration
Group Policy	Centralized configuration and security policies
NTFS Permissions	File and folder access control

Action1	Patch management and endpoint administration
VirtualBox Guest Additions	Virtual machine integration and file sharing

Objectives:

- Deploy an Active Directory Domain Controller
- Create Organizational Units
- Manage user accounts
- Configure Security Groups
- Join Windows clients to the domain
- Implement Group Policies
- Simulate Help Desk support requests (password resets, account unlocks)

Steps Taken:

Step 1: Environment Setup

- Downloaded & installed Oracle VirtualBox
- Downloaded the Windows Server 2022 Evaluation ISO from Microsoft
- Downloaded the Windows 11 ISO for the client machine
- Created a virtual machine for the domain controller

Step 2: Windows Server & Active Directory Installation

- Installed Windows Server 2022 on the virtual machine
- Renamed the server hostname to FL-DC-01
- Restarted the server to apply the hostname change
- Installed the Active Directory Domain Services role using Server Manager
- Promoted the server to a Domain Controller
- Created a new forest with the domain name charlie.local

Step 3: Guest Addition Installation & Account Creation

- Installed VirtualBox Guest Additions and restarted the virtual machine
- Configured a VirtualBox Shared Folder named helpdesklab for file transfer between the host system and virtual machine
- Created 2 new user (Jason & Olive) using Active Directory Users & Computers

Step 4: New Windows Machine & Joining Domain

- Created new Windows 11 machine & installed Windows 11 Pro
- Renamed new machine to “Desktop01”
- Configured static IP addresses and tested connectivity between the Domain Controller and Windows 11 client using ping
- Joined Desktop01 to the charlie.local Active Directory domain and restarted the workstation
- For testing I signed in as user Jason and using the IP of the domain controller connect to the desktop remotely

Step 5: Work Environment Testing & Group Policy

- Disabled user account Jason in AD & deleted Jason’s profile in the local admin account
- Re-enabled Jason then set log in hours to only during work hours
- Set the account to be expired by changing the “End of:” date to the month prior
- Changed some Group Policy like: max password age from 24 to 90 days, min password length from 7 to 12 characters, and account lockout threshold from 1 to 3 attempts
- Simulated an account lockout by entering an incorrect password multiple times and restored access by unlocking the account through Active Directory Users and Computers

Step 6: Patch Management w/ Action1

- Logged into Action1 and connected it to the Server 2022 VM
- Deployed missing updates to the VM through Action1
- Explored the Built-in Reports section and viewed different reports such as patch management, vulnerability management, installed software list.

Step 7: Share Files & Security Groups

- Created a folder named “AdminTech” through Server Manager\File and Storage Services\Shares
- Created sub folders (Tech, HR, Sales) within AdminTech for different groups
- Created a security group named Tech that allows access to \\FL-DC-01\Tech
- Gave Jason and Olive shared access to the Tech folder by making them both members of the Tech security group

Step 8: Changing Settings w/ Group Policy

- I changed the desktop wallpaper on user Jason using a share folder to get a new wallpaper image and Group Policy Management Editor to change the wallpaper from AD
- Configured other settings such as: Screen Saver Timeout > Disabled, Remove Logoff > Enabled, Remove Task Manager > Enabled

Step 9: Action1 AD Deployment

- I installed Action1 Deployer on my Server 2022 VM & deployed an agent via Action1
- I also connected Jason as a new endpoint in Action1

Results:

The Active Directory home lab successfully established a functional Windows Server 2022 domain environment using charlie.local and a Windows 11 Pro client. The environment was used to simulate common IT administration and help desk tasks.

The completed environment provided hands-on experience with several tasks commonly performed by help desk and IT Support technicians, including account administration, access management, workstation configuration, Group Policy management, patch management, and basic troubleshooting.

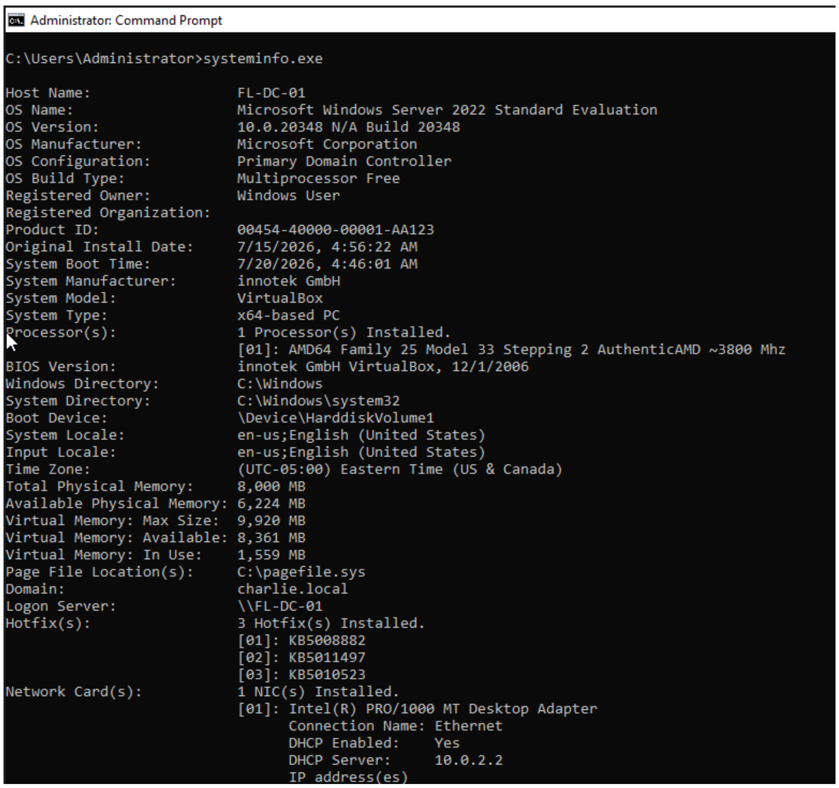
Key Takeaways & Lessons Learned:

Key skills and concepts I developed during the lab include:

- Active Directory: Created and managed users, security groups, domain resources, and account policies.
- Windows Server 2022: Configured and administered a Windows Server environment and Domain Controller.
- Domain Administration: Joined a Windows 11 workstation to an Active Directory domain and verified domain connectivity.
- User Account Management: Practiced password management, account disabling/enabling, account expiration, logon-hour restrictions, and account lockout recovery.
- Group Policy: Created and modified policies to centrally manage security and user experience settings.
- File and Resource Access: Configured shared folders, security groups, and NTFS permissions to control access to network resources.
- Patch Management: Used Action1 to identify and deploy missing Windows updates and review endpoint reports.

- Troubleshooting: Diagnosed and tested network connectivity, domain communication, account access, and policy configuration.
- Remote Administration: Practiced accessing and managing a domain-connected workstation remotely.
- IT Documentation: Documented configuration steps, troubleshooting experiences, and technical results using screenshots and written explanations.

Technical Evidence:

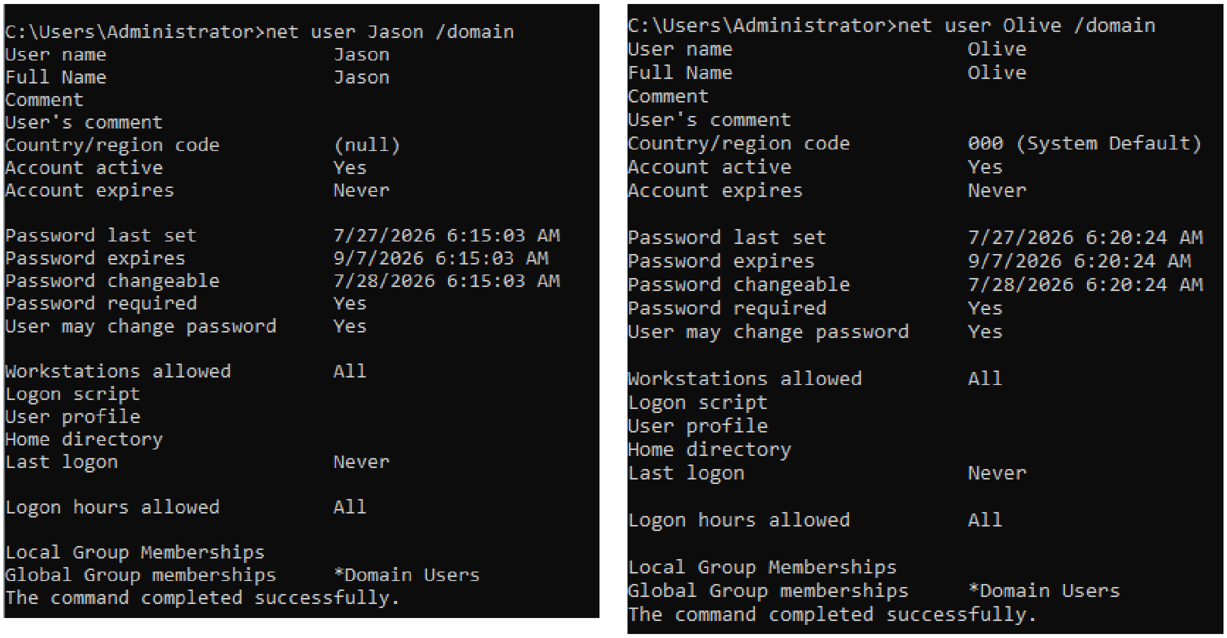


```
Administrator: Command Prompt

C:\Users\Administrator>systeminfo.exe

Host Name:                 FL-DC-01
OS Name:                   Microsoft Windows Server 2022 Standard Evaluation
OS Version:                10.0.20348 N/A Build 20348
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Primary Domain Controller
OS Build Type:             Multiprocessor Free
Registered Owner:          Windows User
Registered Organization:
Product ID:                00454-40000-00001-AA123
Original Install Date:     7/15/2026, 4:56:22 AM
System Boot Time:          7/20/2026, 4:46:01 AM
System Manufacturer:       innotek GmbH
System Model:              VirtualBox
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: AMD64 Family 25 Model 33 Stepping 2 AuthenticAMD ~3800 Mhz
BIOS Version:              innotek GmbH VirtualBox, 12/1/2006
Windows Directory:         C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC-05:00) Eastern Time (US & Canada)
Total Physical Memory:     8,000 MB
Available Physical Memory: 6,224 MB
Virtual Memory: Max Size:  9,920 MB
Virtual Memory: Available: 8,361 MB
Virtual Memory: In Use:    1,559 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    charlie.local
Logon Server:              \\FL-DC-01
Hotfix(s):                 3 Hotfix(s) Installed.
                           [01]: KB5008882
                           [02]: KB5011497
                           [03]: KB5010523
Network Card(s):           1 NIC(s) Installed.
                           [01]: Intel(R) PRO/1000 MT Desktop Adapter
                                 Connection Name: Ethernet
                                 DHCP Enabled:    Yes
                                 DHCP Server:     10.0.2.2
                                 IP address(es)
```

Figure 1: Computer configuration data



```
C:\Users\Administrator>net user Jason /domain
User name                    Jason
Full Name                    Jason
Comment
User's comment
Country/region code          (null)
Account active               Yes
Account expires              Never

Password last set            7/27/2026 6:15:03 AM
Password expires             9/7/2026 6:15:03 AM
Password changeable          7/28/2026 6:15:03 AM
Password required            Yes
User may change password     Yes

Workstations allowed         All
Logon script
User profile
Home directory
Last logon                   Never

Logon hours allowed          All

Local Group Memberships
Global Group memberships     *Domain Users
The command completed successfully.
```

```
C:\Users\Administrator>net user Olive /domain
User name                    Olive
Full Name                    Olive
Comment
User's comment
Country/region code          000 (System Default)
Account active               Yes
Account expires              Never

Password last set            7/27/2026 6:20:24 AM
Password expires             9/7/2026 6:20:24 AM
Password changeable          7/28/2026 6:20:24 AM
Password required            Yes
User may change password     Yes

Workstations allowed         All
Logon script
User profile
Home directory
Last logon                   Never

Logon hours allowed          All

Local Group Memberships
Global Group memberships     *Domain Users
The command completed successfully.
```

Figure 2: New users "Jason" and "Olive" created

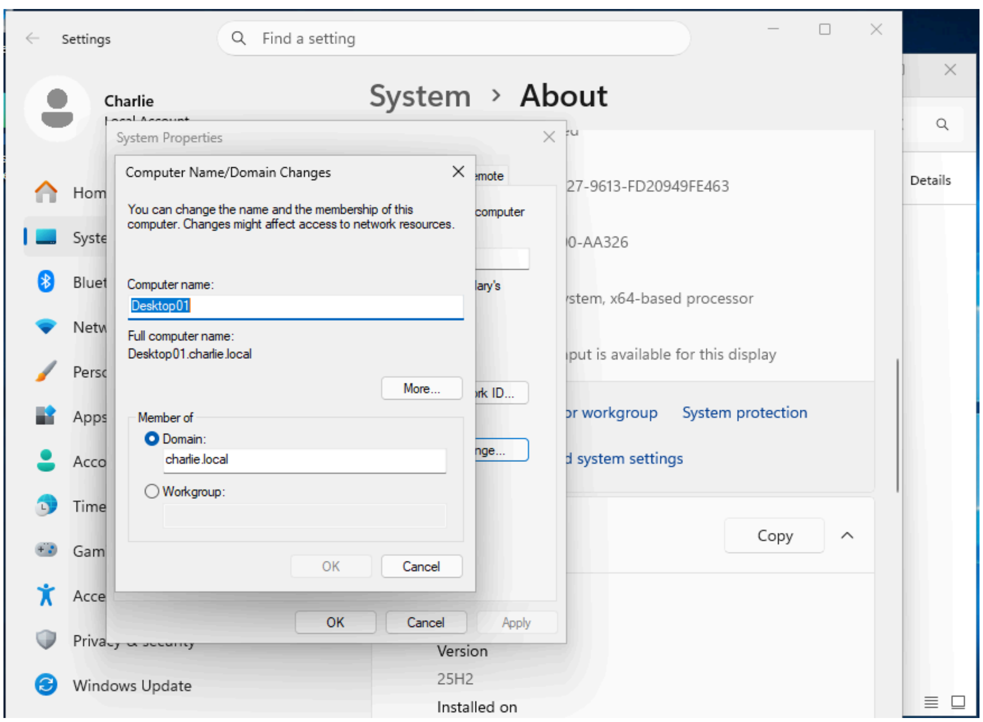


Figure 3: Successfully linked computer to my own domain (charlie.local)

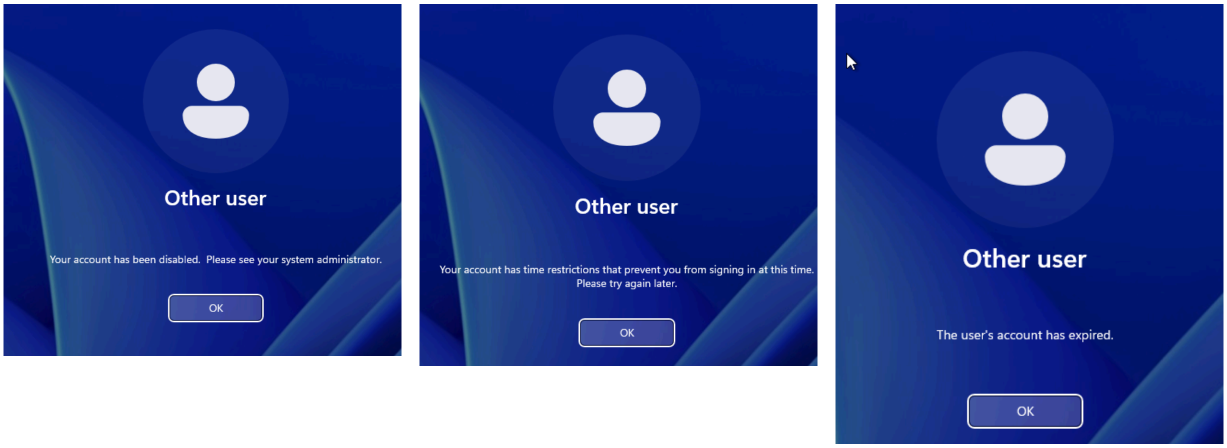


Figure 4: Account “Jason” was disabled through AD, log in hours were changed then sign in was attempted not within work hours, account expire date was changed making the account expired

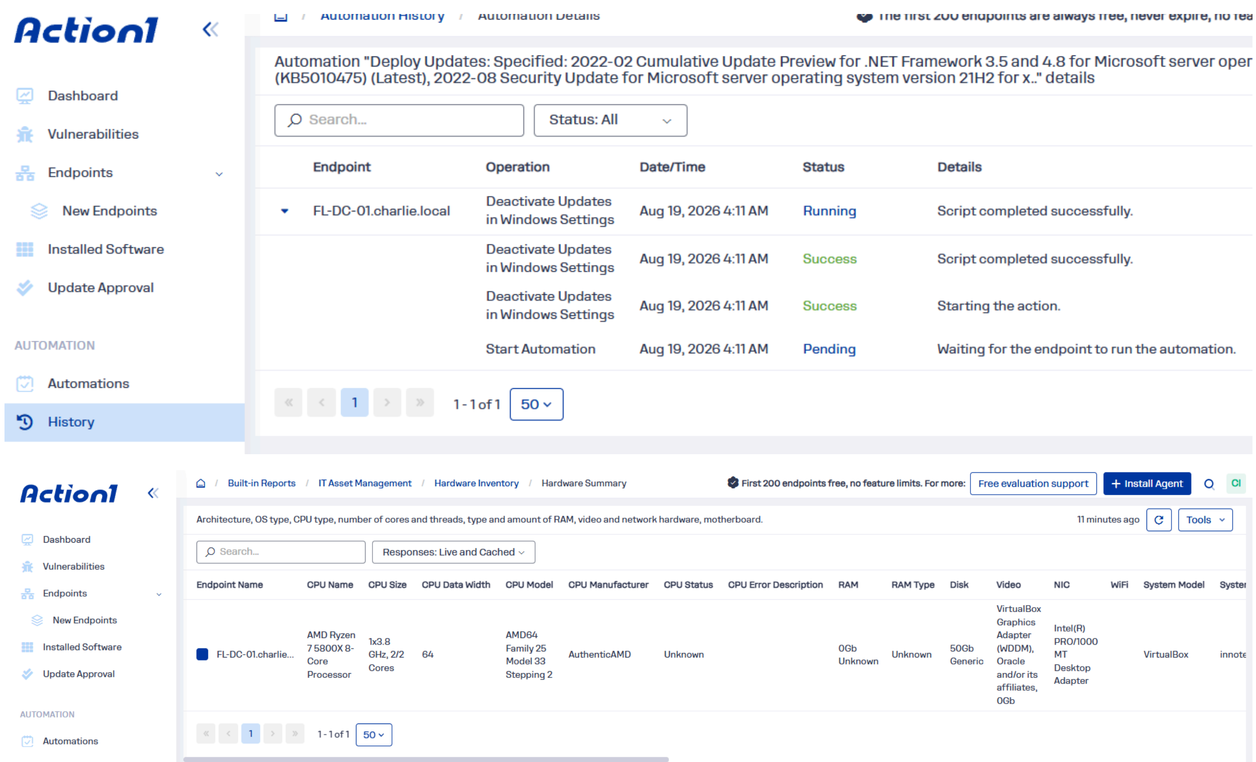


Figure 5: Action1 history list showing the updates that were deployed to the Server 2022 and a full hardware summary from the built-in reports.

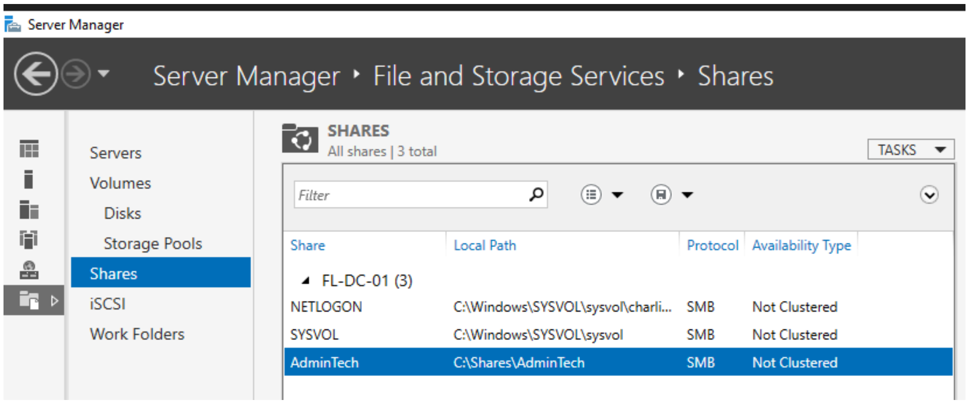


Figure 6.1: Share file “AdminTech” created

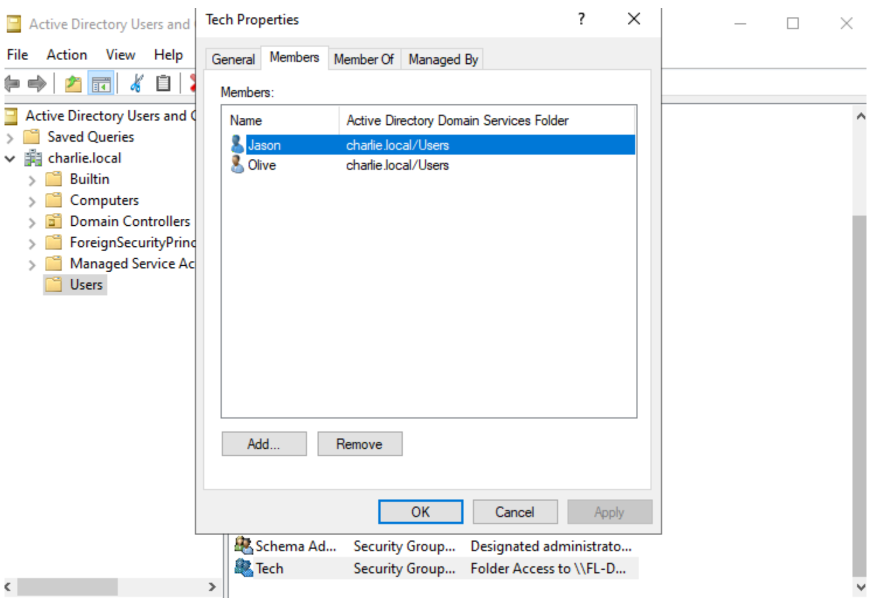


Figure 6.2: Jason & Olive both members of the Tech security group

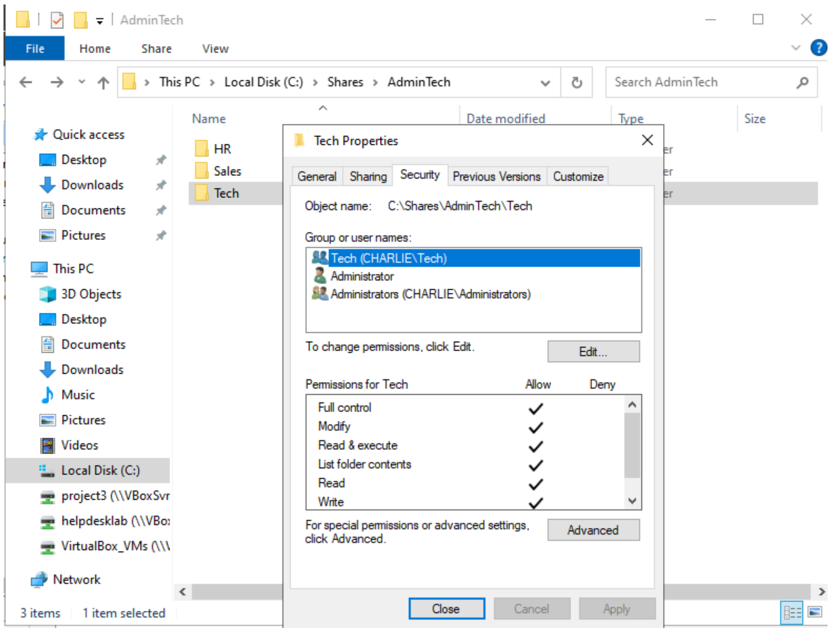


Figure 6.3: Tech security group has shared access to \\FL-DC-01\Tech

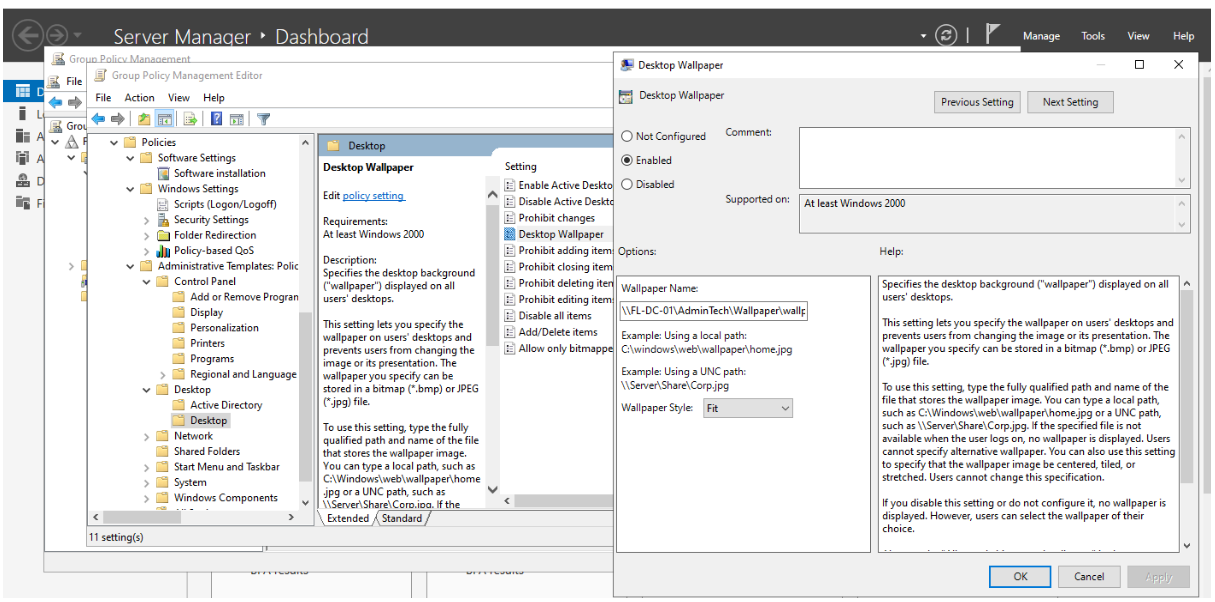


Figure 7.1: Changing the desktop wallpaper to the image from the share folder in the Group Policy Management Editor

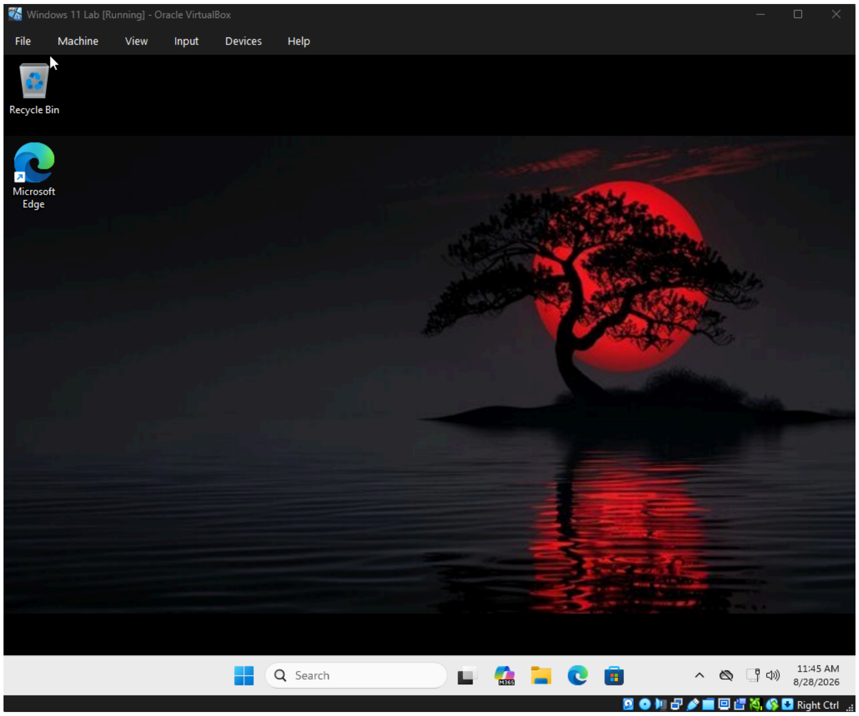


Figure 7.2: Jason's desktop wallpaper result after the change

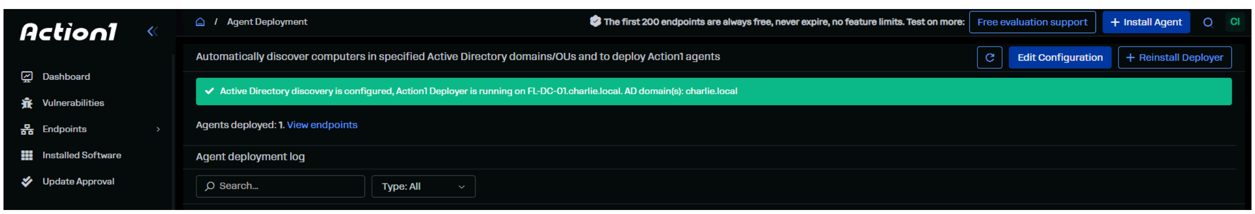


Figure 8.1: The Action1 Deployer agent running on charlie.local

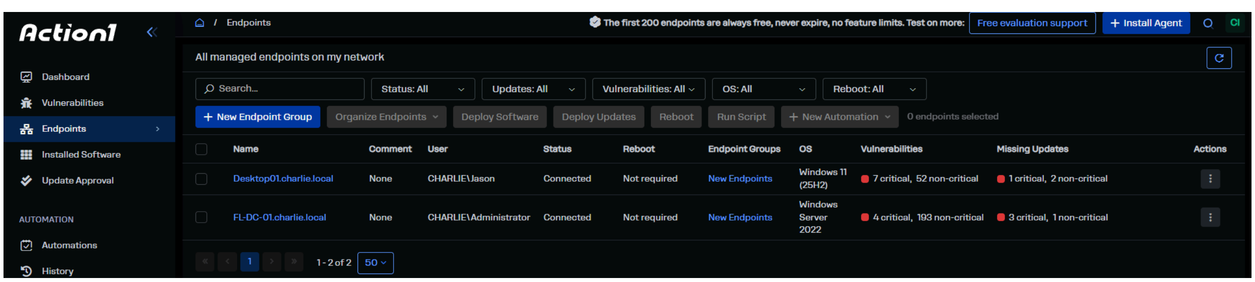


Figure 8.2: Jason is also connected to Action1